Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2625519

Meritocracy vs. Popularity: comparison and Redesign of DWTS Mechanisms

Summary

To reach a balance between professional meritocracy and mass entertainment in the *Dancing with the Stars* evaluation system, this paper presents a robust analytical framework that transitions from data reconstruction to systemic redesign.

First, facing the challenge of "black-box" fan voting data, we develop an **Inverse Inference Model** utilizing **Monte Carlo simulation** coupled with **Bayesian updating**. By treating historical elimination results as constraints, we reconstruct the posterior distribution of latent fan votes. Our model achieves a high reconstruction stability with a median **Coefficient of Variation (CV) of 0.293**, significantly below the high-volatility threshold (0.5), providing a rigorous empirical foundation for subsequent causal analysis.

Second, Using the reconstructed data, we employ a **dual-branch Linear Mixed-Effects Model (LMM)** to decouple the multi-dimensional factors influencing competition outcomes. By isolating professional dancers as random effects, we quantify the marginal contributions of celebrity attributes. Findings reveal a profound **"Preference Divergence"**: judge scores are strictly merit-driven (demonstrating significant negative correlation with age), whereas fan votes are narrative-driven, showing extreme positive bias toward contestants with "Reality TV" backgrounds.

Third, We conduct **counterfactual simulations** to diagnose the mathematical vulnerabilities of historical aggregation methods. Quantitative results show that the **Percentage System** exhibits a significantly higher Spearman correlation with fan votes (**0.8891**) compared to the **Rank System** (**0.8137**), explaining why popularity can easily overwhelm technical merit in percentage-based seasons. We further demonstrate the introduction of a **Judges' Save** mechanism acts as a true "circuit breaker," raising the **Skill Protection Index (SPI)** from 50.5% to **88.6%**.

Ultimately, we propose the **Dynamic Adaptive Scoring System**. This system integrates **Entropy-TOPSIS** for real-time, objective weighting—automatically adjusting the influence of judges based on the weekly differentiation of performances—with a **Quantitative-based Judges' Save** protocol triggered by a controversy threshold. Our final evaluations demonstrate that DASS achieves a balance: it maximizes professional integrity by maintaining an **SPI above 92%** while preserving a **Fan Influence Index (FII)** within a commercially viable range (48%-51%).

Contents

1 Introduction

1.1 Problem Background

The operational core of *Dancing with the Stars* (DWTS) lies in the dynamic interplay between two distinct forces: **Judge Scores**, representing professional merit and technical precision, and **Fan Votes**, representing popularity and mass entertainment value. This system is not merely an aggregation of scores but a mathematical negotiation between meritocracy and democracy.

Historically, the show has oscillated between two aggregation methods—**Rank-based** and **Percentage-based**—to mitigate "unfair" outcomes where technically inferior but popular contestants (e.g., Bobby Bones) secure victories. However, a critical challenge exists: the **Fan Vote** is a strictly guarded secret (a latent variable). Consequently, any evaluation of historical fairness or design of future rules lacks direct empirical foundation. The essence of this problem, therefore, is to deconstruct and redesign a complex social evaluation system under conditions of censored data.

1.2 Clarifications and Restatements

To address mentioned challenges, we reframe the problem into four tightly coupled mathematical tasks:

Task1: Reconstruction of **Latent Variables.** Faced with unobservable fan votes, we must construct an Inverse Inference Model. Operating on a weekly basis, we will use the known "Judge Scores" and the binary "**Elimination** Results" as constraints to reconstruct the probability distribution of fan votes. Crucially, we aim to go beyond point estimates by performing Uncertainty Quantification, providing confidence intervals or percentage distributions of the rankings as the foundation for all subsequent analyses.

Task 2: Deconvolution of Multidimensional Influence Factors. We need to establish statistical models to quantify and strip away the marginal contributions of different factors: first, the **objective attributes** of the contestants. The second is **subjective performance** (judges' ratings and fans' wishes). We need to explore how these characteristics differentially affect the judges' judgments and the voting behavior of fans.

Task 3: Comparative Dynamics of Aggregation Mechanisms Using the reconstructed data, the differences between the "ranking" and "percentage" aggregation algorithms in dealing with the differences between judges and fans are comparatively analyzed. The susceptibility of these two mechanisms to extreme cases is examined, and the modifying effect of introducing the "Judges' Save" mechanism (i.e., the judges decide the bottom two eliminators) on the final ranking and the fairness of the competition.

Task 4: Multi-objective Optimization and Multi-objective Optimization Based on the above analysis, we will design a **new adaptive scoring system**. This is not a single-dimensional solution, but a multi-objective optimization problem. Additionally, we are supposed to give reasons about the strength of the new mechanism

1.3 Our Work

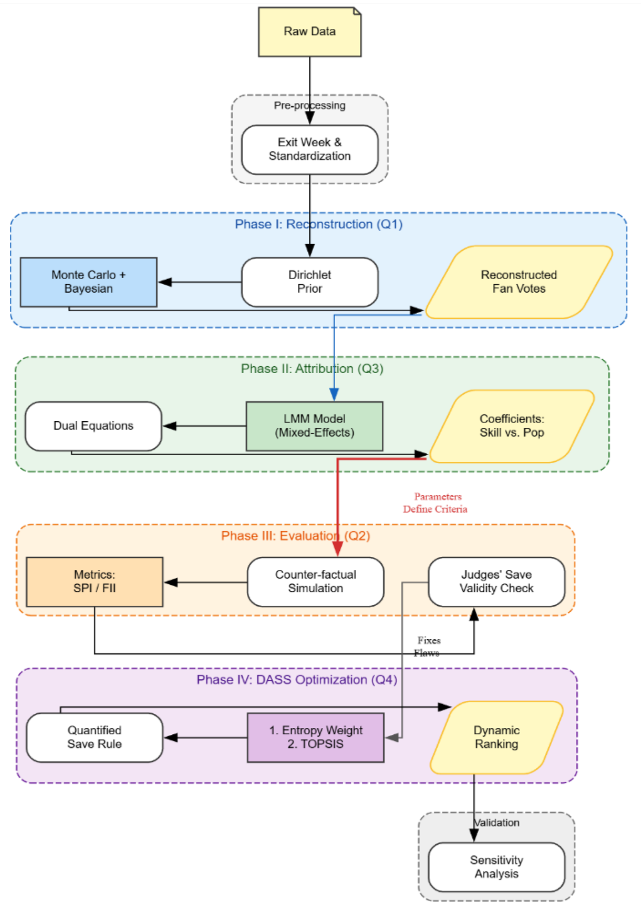


Figure1: Our Work

2 Preparation for Modeling

2.1 Assumptions and Justifications

To ensure the validity and tractability of the model, the following assumptions are established:

Assumption of Voting Distribution: We assume that the fan voting vector $V_t = [v_1, \ldots v_{N_t}]^{\mathrm{T}}$ follows a symmetric Dirichlet distribution. This implies that, in the absence of external influences, the probability of any contestant receiving a specific proportion of votes is uniform. This assumption justifies the approximation using a normal distribution, as the competition outcome is determined by the proportion of public votes rather than the absolute vote count. **Adherence to Elimination Rules:** It is assumed that the production team strictly adheres to the publicized ranking or percentage-based rules for contestant elimination. We assume there are no undisclosed manipulations or irregularities affecting the administrative process. **Proportionality of Outcomes:** We assume that the competition results depend solely on the share (proportion) of votes received and are independent of the absolute magnitude of the total votes cast. **Independence and Integrity:** We assume that the weekly judges' scores and fan votes are mutually independent variables. Furthermore, we assume the absence of malicious behavior, such as strategic scoring by judges or coordinated malicious voting by the audience

2.2 Notations

Symbol	Definition & Description
t	Index representing the competition week (t = 1, 2...T)
i	Index representing the contestant(i= 1, 2,...,N_t)
N_t	Number of active contestants remaining in week t
$J_{i,t}$	The judge's raw score awarded to contestant i in week t
$V_{i,t}$	The latent fan vote share (percentage) for contestant i
V_t	The vector of fan vote shares,V_t= $[v_1,\dots v_{N_t}]^{\mathrm{T}}$
S_t	The set of surviving contestants in week t
ϵ_t	The set of eliminated contestants in week t
$Score_{i,t}$	Aggregate score calculated via Rank or Percentage method
Ω_t	The feasible region of fan votes on the standard simplex
$Dir(\alpha)$	Dirichlet prior distribution with parameter vector α
α	Hyperparameter controlling the sparsity of the prior
$CI_{i,t}$	Certainty Index quantifying the estimation confidence
N_{total}	Estimated total number of fan votes based on viewership
$Y_{ijk}^{(J)}, Y_{ijk}^{(F)}$	The actual score of the i-th celebrity led by the j-th dance partner in week k
β_0	Intercept term
β, δ, λ	Coefficients of different features
$u_k^{(J)}, u_k^{(F)}$	The true ability gain of the dance partner
ϵ_{ijk}	Random error term
Δ_R	the degree of dispute

CV	Coefficient of Variation

2.3 Data Preprocessing

2.3.1 For retired players

There are situations where players withdraw from the competition midway to a certain week, respectively in season 3, 7, 9, 16, 18, 21, 28, 29,31 withdrawals. We looked for the results of each round of retirees and compared them with the average judges' scores of all players in each round and found that the results of retired players were generally lower than the average level of participants, so we took the first week when their lack of results as Exit Week.

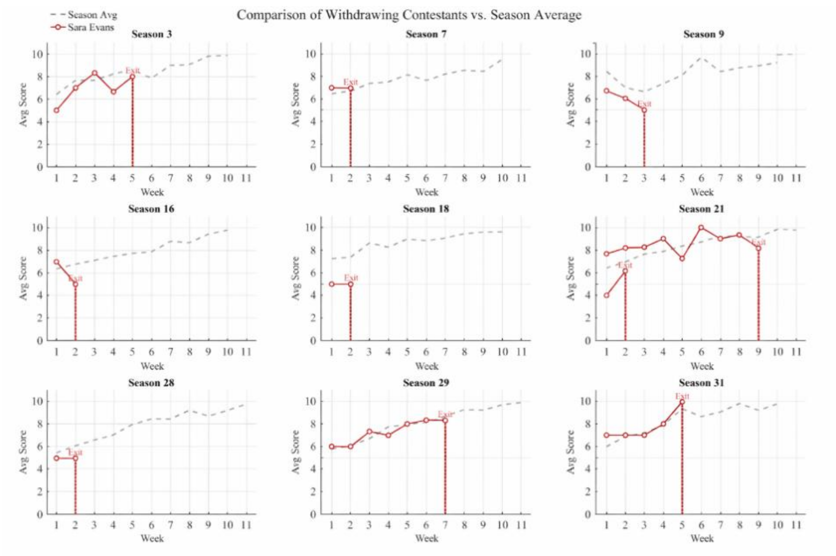
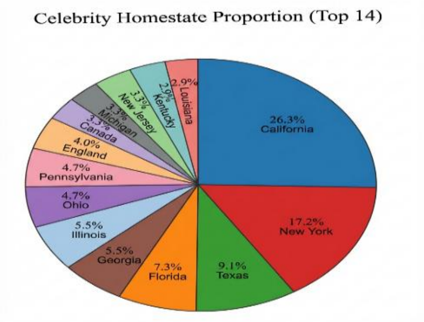


Figure2: Comparison of the results of the eliminated players

With respect to celebrities, considering that their inherent characteristics may lead to varying degrees of favorability or bias from judges or fans, we therefore calculate the frequency of celebrities' industries and home states. This enables subsequent analyses of how different professions and home states of celebrities may generate inherent impressions among judges and fans



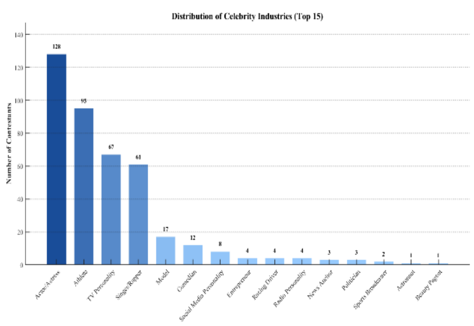


Figure3: Celebrity Home states Proportion Figure4: Distribution of Celebrity Industries

With respect to dancers, in order to analyze in subsequent sections the impact of different dancers on celebrity performances, we calculate the frequency of dancers' appearances and assign corresponding indicator values to them.

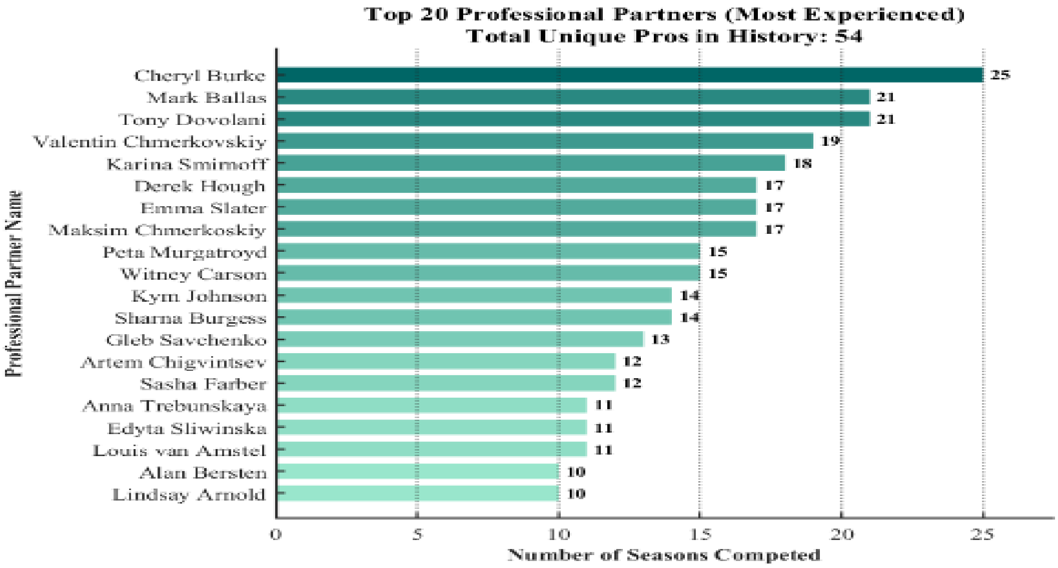


Figure5: distribution of dancers

3 A Three-Stage Inversion Model for Fan Vote Reconstruction

3.1 Introduction of the Three-Stage Inversion Model

To infer the latent variables (fan votes) from censored observations—where only elimination outcomes are available—we develop a three-stage inversion model consisting of an *a priori* hypothesis, Monte Carlo simulation, and posterior filtering.

3.2 Construction of a Prior Distribution

3.2.1 Assumptions of Dirichlet's distribution

Since fan votes represent proportions that sum to 1 (or 100%), with each contestant receiving a strictly positive share, the Dirichlet distribution provides a natural and principled way to model this multidimensional compositional relationship.

Assuming there are n players in week t, the fan vote distribution vector is $V_{fan}= (v_1, v_2, ..., v_n)$. We use the judges' normalized scoring as a priori expectations for fan voting, which means we assume that fans converge with the judges' aesthetics to some extent, but there is a random deviation. Since the parameter vector of the Dirichlet distribution, α correlates with the accuracy of the model, we construct the component vectors to determine the optimal parameter α.

3.2.2 Determination of Parameters in the Symmetric Dirichlet Distribution

The parameters in the Dirichlet distribution α correlates with the accuracy of the Monte Carlo simulation, in order to determine this parameter. We build dynamic expressions about α to accommodate changes in the judges' scores from week to week.

(1) Definition of the popularity score

The Career Heat Score is defined asS_{ind}To avoid two-level differentiation, we perform logarithmic smoothing processing to map the scores to the [0,1] interval.

The Region Bonus Score is defined as: We quantify it withS_{loc} the regional frequencies shown in the Location_Rank_ChartS_{loc}, so there is

$$S_{loc}=\frac{Frequency}{\max(Frequency)}$$

Age advantage score is defined as $S_{age}$to quantify the possible impact of a player's age. We assume that in general, the younger the contestants, the more energetic they are, and they are more likely to catch the fancy of the audience.

$$s_{age}=1-\frac{Age-\min\,(Age)}{\max(Age)-\min\,(Age)}$$

Based on the construction of the above metrics, we define the popularity score as$S_{pop}$where is the weight of the corresponding indicator$\omega1$，$\omega2$，$\omega3$

$$S_{pop}=\omega_1 S_{ind}+\omega_2 S_{loc}+\omega_3 S_{age}$$

(2) Standardization of judges' scoring

Standardize the results of the weekly judges' scores. Among them, i_{scores} is the average score of the judges in week i of the surviving partner, and $\max(i_{scores})$ is the maximum score of the judges in the week i of the surviving partner.

$$j_{norm}=\frac{i_{scores}}{\max\,(i_{scores})}$$

(3) Construct α expressions

α=1+$K_1\times j_{norm}$+ $K_2\times S_{pop}$

We use the iterative method to finally obtain ω_1, ω_2, ω_3, K_1, K_2 the best αparameters when the weights are 0.8, 0.1, 0.1, 5.5 and 4.0, respectively, that is, the Monte Carlo simulation method has the highest accuracy.

$$V_{fan}\sim Dirichlet(\alpha_1,\alpha_2,\dots\alpha_n)$$

3.3 Application of Monte Carlo Simulation Method

3.3.1 Stage Separation

Across Seasons1–34, we partition the Monte Carlo simulation framework into three phases to reflect the show's evolving vote-combination rules:

Stage1: Season1-2 (based on ranking)

Stage2: Season3-27 (based on percentage)

Stage3: Season28-34 (based on ranking and guaranteed mechanism)

The voting combination method is the same as that in Stage 1, but the addition of the guaranteed mechanism is equivalent to introducing elimination probabilities rather than direct elimination.

3.3.2 Procedure of the Monte Carlo Randomized Simulation

We employ Monte Carlo methods to conduct large-scale randomized trials. For each week, we run 100,000 independent simulations:

(1) Sampling: Draw a random vector from the Dirichlet distribution to represent the hypothetical fan-vote proportions in the i-th simulation, denoted $v_{fan}^{(1)}$.

(2) Synthesizing: Combine the sampled fan votes with the observed judges' scores to compute the composite score according to the season-specific aggregation rules.

(3) Ranking: Rank contestants by the composite score and identify the "simulated eliminated" contestant under the i-th simulated scenario.

3.4 Bayesian Updating and Posterior Filtering Results

The simulation outcomes are compared with the observed elimination results [2] to assess model validity:

Situation A (Match): If the predicted eliminated contestant coincides with the actual eliminated contestant, the corresponding fan-vote data are considered sufficiently credible and are therefore retained.

Situation B (Mismatch): If the predicted eliminated contestant differs from the actual eliminated contestant, the fan-vote data are deemed unreliable and are consequently discarded. The final simulation outcomes are presented in Figure 6.

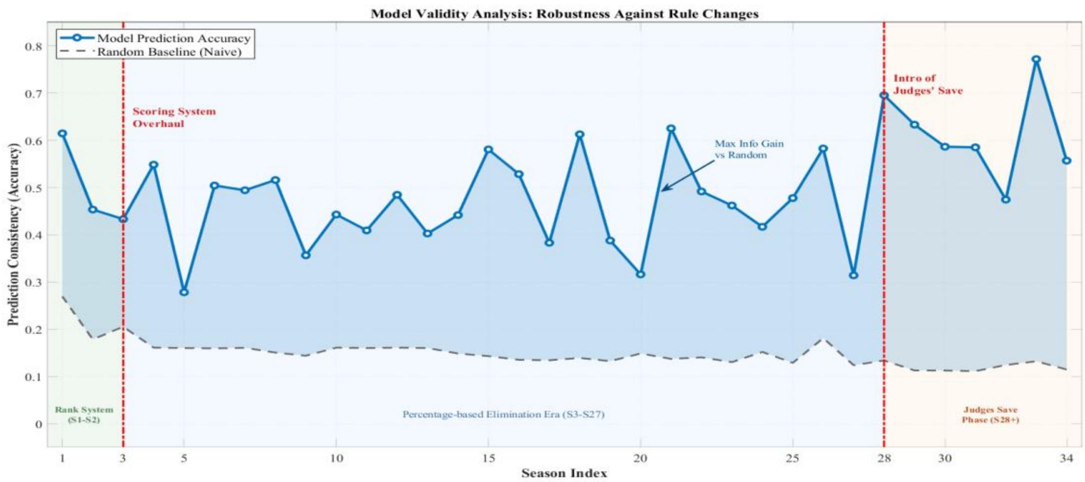


Figure6: Model Validity

From these results, we observe that:

(1) In Season 28, the performance of the simulation model exhibits an abrupt change, with accuracy exceeding the average level observed in the preceding season.

(2) When the parameter α is estimated based on weekly judges' score outcomes, the simulation achieves significantly higher accuracy than simulations in which α is assigned randomly.

3.5 Measurement of Model Certainty

We define the coefficient of variation, denoted by CV_i, to quantify the uncertainty in the total number of fan votes. Let S_i be the standard deviation of the scores of the remaining contestants in week i, and let N_i be their mean score in week i. We compute

$$CV_i = \frac{S_i}{N_i}$$

We calculate the average CV values for the same week in each season, as shown in the figure below:

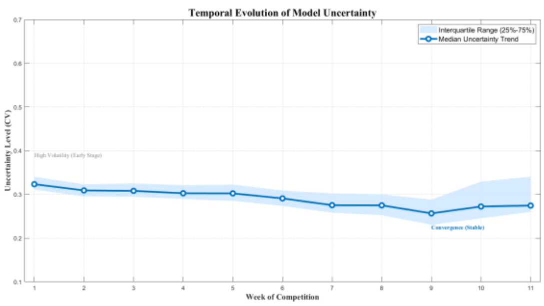


Figure7: Temporal Evolution of Model Uncertainty

In the line chart, the light-blue shaded band represents the interquartile range (IQR, 25%–75%) of the seasonal uncertainty coefficients. In principle, a narrower band indicates convergence of the model, meaning that CV_i varies only slightly across adjacent weeks. Moreover, the median global uncertainty is 0.293 (below 0.5), which is well under the statistical threshold commonly associated with "high volatility." Therefore, the certainty of the total fan-vote estimates remains essentially stable under small week-to-week variations.

4 Deconvolution of Multidimensional Influence Factors

4.1 Data Processing and Feature Engineering

To quantify contestant popularity, we introduced a new metric: Partner Experience, defined as the cumulative number of appearances a professional dancer has made in previous seasons. We posit that a higher number of appearances implies a more seasoned partner, which may positively correlate with popularity. Before modeling, we standardized continuous variables—including age, partner experience, and judges' scores—to facilitate convergence and interpretation. Crucially, given the data structure, we treat Partner Experience (proxy for the professional dancer's individual impact) as a random effect, while modeling all celebrity characteristics (e.g., occupation, region) as fixed effects.

4.2 The Dual-Perspective Linear Mixed-Effects Model (LMM)

4.2.1 *Formulation* of Dual Equations

Since different celebrities across seasons may share the same professional dance partner, the performance outcomes are inherently **nested and correlated**. To account for this dependency and strictly isolate the celebrity's influence, we developed a **Dual-Perspective Linear Mixed-Effects Model**.

We established two parallel LMMs to capture evaluation criteria from the **"Professional Perspective" (Judges)** and the **"Public Perspective" (Fans)**, respectively. The rationale for employing a mixed-effects framework is the **nested structure** of the DWTS dataset: multiple stars are grouped under the same professional partner. By modeling the dance partner as a **Random Intercept**, we effectively filter out the interference of the partner's individual ability/popularity. This ensures that the remaining variance allows us to quantify the influence of the celebrity's own characteristics more precisely.

$$Y_{ijk}^{(J)} = \beta_0 + \beta_1 X_{\text{Age}} + \beta_2 X_{\text{Exp}} + \sum_m \beta_m X_{\text{Ind}} + \sum_n \delta_n X_{Reg} + \underbrace{u_k^{(J)}}_{\text{Partner RE}} + \epsilon_{ijk}$$

$$Y_{ijk}^{(F)} = \gamma_0 + \gamma_1 X_{\text{Age}} + \gamma_2 X_{\text{Exp}} + \sum_m \gamma_m X_{\text{Ind}} + \sum_n \lambda_n X_{Reg} + \underbrace{v_k^{(F)}}_{\text{Partner RE}} + \epsilon'_{ijk}$$

Among them, X_{Age}, X_{Exp}, X_{Ind} celebrity feature vectors, u_k, v_k, capture the random effects of individual dance partners, $Y^{(J)}$, $Y^{(F)}$, Standardized judge scoring and fan turnout

4.2.2 Model Architecture and Random Effects

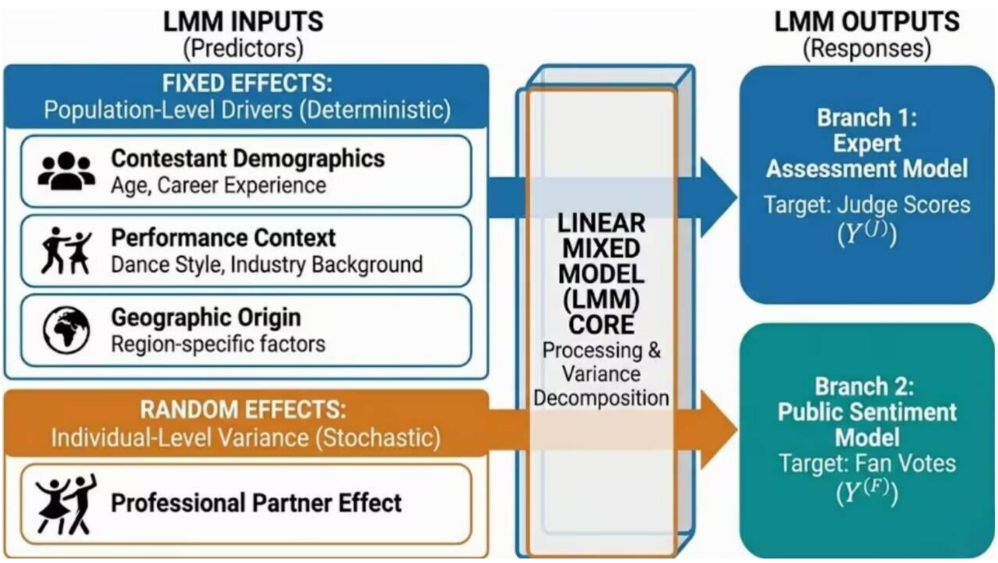
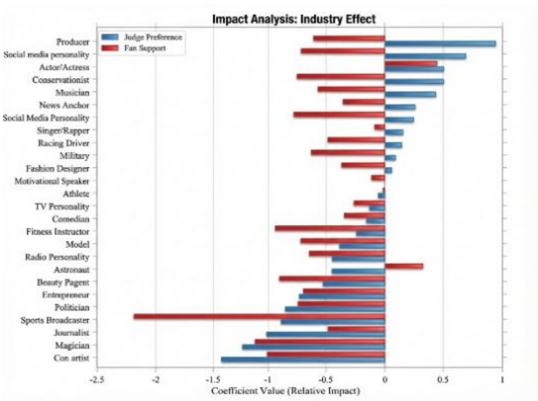
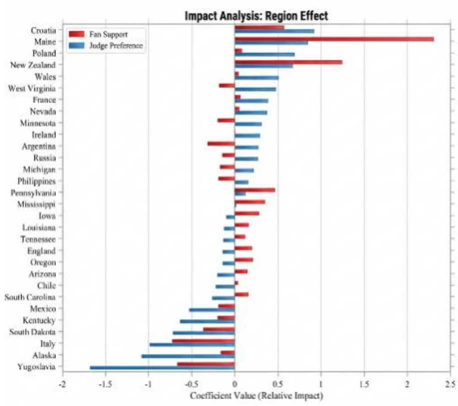


Figure 8: System Architecture of the Dual-Perspective Linear Mixed-Effects Model

4.2.3 Interpretation of Fixed Effect Coefficients

Based on the LMM, we estimated the coefficients for each celebrity characteristic. The results are summarized in the table below:





(a) Divergence in Occupational Impact (b) Evaluation of Regional Influence

Figure 9: Comparative Analysis of Fixed Effects on Judge and Fan Preferences

Among them, β_{Judge} and β_{Fan} represent the **fixed-effect coefficients** corresponding to celebrity characteristics for judges' scores and fan votes, respectively. These coefficients quantify the magnitude of favorability from each perspective. $Diff$ denotes the discrepancy between the two ($\beta_{Fan} - \beta_{Judge}$). Through the analysis of the coefficients and model output, we draw the following key conclusions:

(1) Impact of Occupation:

Reality TV stars often emerge as generators of controversy. For this demographic, β_{Fan} is significantly positive and high, whereas β_{Judge} is close to zero or even negative. This discrepancy suggests that a large inherent **fanbase** translates "social capital" into a high volume of votes, yet their dance skills often lack the technical rigor to impress professional judges. This empirically proves that fan voting is heavily driven by pre-existing popularity and entertainment value rather than strictly by performance quality.

In contrast, **Athletes** typically possess both physical prowess and public appeal. Both their β_{Judge} and β_{Fan} are positive, with β_{Judge} consistently ranking the highest among all occupations, indicating a balance of skill and popularity.

(2) Impact of Region:

Regional differences exhibit negligible impact on competition outcomes. The coefficients β_{Judge} and β_{Fan} for contestants from different regions lacked any discernible trend. This suggests that the show's fan voting behavior is driven by **national media exposure** rather than local geographic biases.

(3) Impact of Dance Partners:

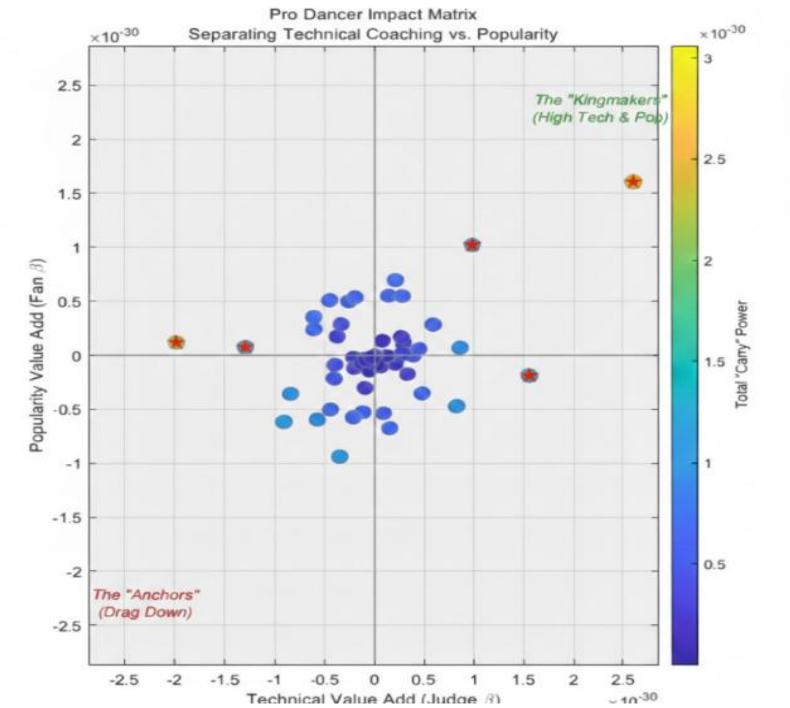


Figure10: Pro Dancer Impact Matrix

As illustrated in Figure 10, the distribution of the estimated effects corresponding to **partner experience** is highly concentrated with small numerical magnitudes. This indicates that, after controlling for other factors, the specific popularity or experience of the professional dancer has a minimal direct impact on the celebrity's final performance metrics.

5 Comparative Dynamics And Counterfactual Reconstruction

5.1 Mathematical Formalization of Mechanisms

To rigorously compare the fairness of history voting rules, we mathematically formalized the two primary mechanisms used in DWTS:

Method 1: The Points Ranking Method

In this system, the raw scores are converted into ordinal ranks. Let R_{Fan} be the rank derived from fan votes and R_{Judge} be the rank derived from judges' scores (where Rank 1 denotes the highest/best performance). The composite score S_1 is defined as:

$$S_1 = R_{Fan} + R_{Judge}$$

The final ranking ($Rank_1$) is determined by the **ascending order** of S_1 (i.e., a lower sum indicates a better placement).

Method 2: The Percentage Method

In this system, the raw magnitude of votes is preserved. Let P_{Fan} and P_{Judge} represent the proportional share of fan votes and judges' scores for contestant i, respectively:

$$P_{Judge} = \frac{J_i}{\sum J_k}, P_{Fan} = \frac{V_i}{\sum V_k}$$

The total proportion S_2 is calculated as:

$$S_2 = P_{Judge} + P_{Fan}$$

The final ranking ($Rank_2$) is determined by the descending order of S_2 (i.e., a higher proportion indicates a better placement).

5.2 Statistical Evaluation of Aggregation Methods

5.2.1 Bias Detection via Spearman Rank Correlation

To quantify the inclination of each mechanism towards either fan preference or professional judgment, we utilized the **Spearman Rank Correlation Coefficient (ρ)** We defined four correlation indicators to measure the consistency between the final outcomes and the input preferences:

Ranking Method Consistency*:*

$$\rho_1 = Spearman(Rank_1, R_{Fan}), \rho_3 = Spearman(Rank_1, R_{Judge})$$

Percentage Method Consistency*:*

$$\rho_2 = Spearman(Rank_2, R_{Fan}), \rho_4 = Spearman(Rank_2, R_{Judge})$$

Results:

The calculation yields $\rho_1 = 0.8137$ and $\rho_3 = 0.8127$, indicating that the **Ranking Method maintains a balanced influence** between fans and judges.

In contrast, for the Percentage Method, $\rho_2 = 0.8891$ (extremely high fan correlation) while $\rho_4 = 0.6478$ (significantly lower judge correlation). This mathematically proves that **the Percentage Method is heavily skewed towards fan voting**, effectively suppressing the judges' influence.

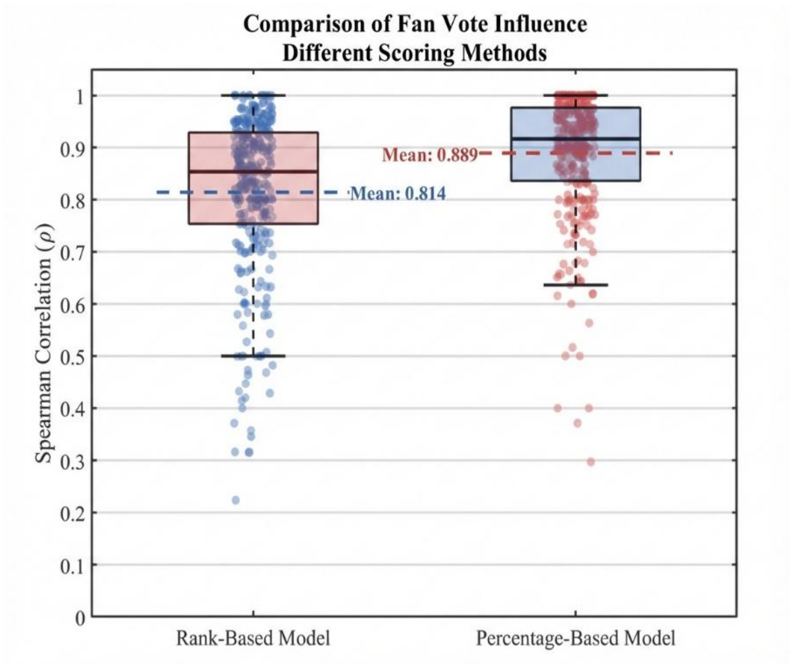


Figure 11: Boxplot Analysis of Spearman Correlations (ρ) between Fan Votes and Final

Outcomes. The percentage-based mechanism (right) exhibits a significantly higher mean correlation coefficient (ρ≈0.889), in contrast to the rank-based mechanism (left) with a mean correlation coefficient of ρ≈0.814. This result confirms that the percentage-based mechanism is more biased toward fan voting.

5.2.2 Variance Disparity Analysis: The Dominance of Fan Voting

To thoroughly elucidate the mathematical nature of why the percentage system is skewed towards fan interests, we introduced **Variance Analysis**. We calculated the standard deviation of fan votes and judges' scores independently for each week across all contestants. Our analysis revealed a consistent inequality: in the vast majority of cases, the dispersion of fan votes was significantly greater than that of judges' scores (i.e., $\sigma_{Fan} \gg \sigma_{Judge}$). This highlights a fundamental structural difference between the two mechanisms:

The **Percentage System** directly preserves this massive variance in fan voting, allowing the raw magnitude of popularity to dominate the final metric. In contrast, the **Ranking System** acts as a **"Variance Compressor."** It forces the vast disparities in fan votes to be compressed into a linear arithmetic sequence (e.g., 1, 2, 3...) with uniform intervals.

Consequently, the Ranking System artificially eliminates the "scale advantage" of the fan base, neutralizing the mathematical imbalance caused by the sheer magnitude difference between the millions of fan votes and the bounded 1-10 scale of judges' scores.

5.2.3 Case Study Counterfactual Simulation

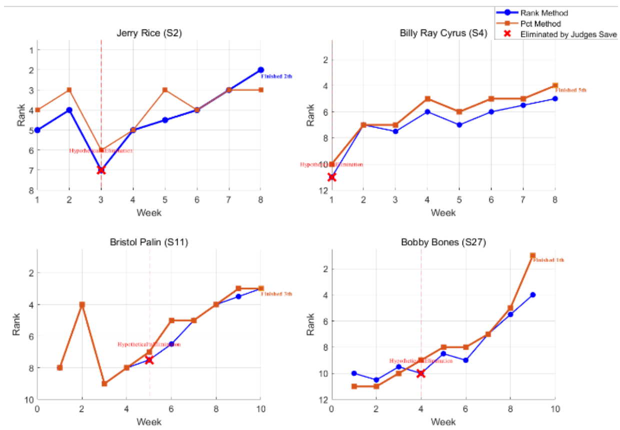


Figure 12: Counterfactual Simulation of Historical Extremes: Impact of Mechanism

To empirically measure the impact of different mechanisms on extreme outliers, we selected four historically controversial contestants for reconstruction. We specifically focus on the **"Bobby Bones Paradox" (Season 27)** as a primary case study.

The Historical Reality: Under the Percentage System, Bones won the championship despite consistently receiving the lowest scores from judges. His massive fan variance mathematically overwhelmed the professional critique, creating a widely criticized outcome.

Counterfactual 1: Switching to the Rank System: Our simulation reveals that under the Rank System, Bones' final placement would drop from 1st to 4th. While the Ranking System mitigates the "winner-takes-all" effect of massive vote counts, Bones would still survive until the semi-finals. His fan base was sufficiently large to prevent him from falling into the bottom rank, even when vote magnitudes were compressed.

Counterfactual 2: Introducing the "Judges' Save": When applying the "Judges' Save" mechanism, Bones triggers the elimination threshold in Week 6. As the lowest-ranked contestant by professional criteria (Judges' Rank), he would have been eliminated immediately, fundamentally altering the trajectory of Season 27 in favor of dance proficiency.

On the one hand, we can observe from the graph that in most cases, the controversial celebrities used for analysis under the percentage method end up ranking higher than those used in the points ranking method. Because these four controversial celebrities generally have low judge scores and relatively high fan votes, we derive that the percentage method expands the influence of fan votes on the final result compared to the points ranking method.

On the other hand, if the Judges' Save mechanism is used, these controversial celebrities will be eliminated early in the corresponding season. However, The truth is that they survive until the later stages of the season and even get a higher final ranking. This proves that the Judges' Save mechanism greatly reduces the number of cases where participants achieve better results based solely on fan popularity. At the same time, this mechanism also protects players who are less popular among fans but actually perform well.

5.2.4 Fairness Quantification via Novel Metric Indices (SPI & FII)

To objectively evaluate the "Fairness vs. Popularity" trade-off, we constructed two novel indicators: the **Skill Protection Index (SPI)** and the **Fan Influence Index (FII)**.

(1) Construction of Metrics

Let N_W be the number of contestants in Week W. We define the set of contestants as $C = \{1,2,\dots,N_W\}$. Let J_i denote the judges' score and F_i denote the estimated fan votes for contestant i.

We define two baseline events:

Technical worst event: E_{skill}= { i | J_i= min (J_k) k∈C}

The least popular event: E_{fan}= { i | F_i=min (F_k) k∈C}

For the three different mechanics mentioned above, we use simulations to get the eliminators of the week $x_{elim}^{(M)}$.

Then, the evaluation indicators Skill Protection Index (SPI) and Fan Influence Index (FII) are evaluated

$$\text{SPI（M）} = \frac{1}{T}\sum_{t=1}^{T} ||(x_{elimt}^{(M)} \in \mathrm{E}_{\text{skill,t}}) \times 100\%$$

$$\text{FII（M）} = \frac{1}{T}\sum_{t=1}^{T} ||(x_{elim,t}^{(M)} \in \mathrm{E}_{\text{fan,t}}) \times 100\%$$

Among them,

High SPI: Indicates the mechanism efficiently eliminates contestants with low professional skills (favors **Meritocracy**).

High FII: Indicates the mechanism efficiently eliminates contestants with low popularity (favors **Audience Influence**).

(2) Results and Interpretation

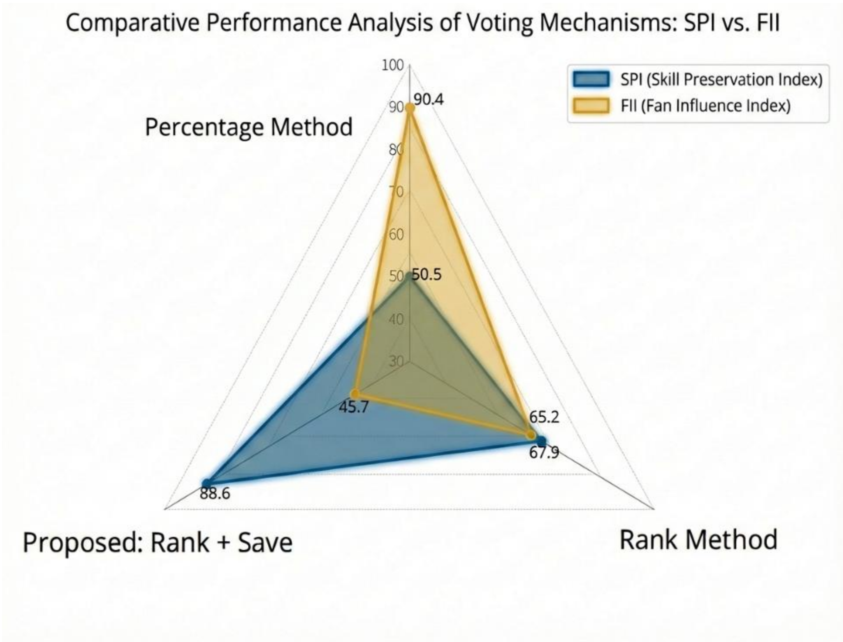


Figure 13: Multi-Criteria Performance Comparison via Radar Analysis: SPI vs. FII.

As illustrated in the graph, under the **Percentage Mechanism**, the Fan Influence Index (FII) reaches approximately **90.4%**, while the Skill Protection Index (SPI) is merely **50.5%**. This statistical disparity indicates that competition outcomes are predominantly driven by fan preference. Professional skills, as reflected in performance evaluations, cease to be the dominant factor in elimination decisions in this scenario. It further suggests that the subjectivity of fan voting significantly outweighs the objective assessment of the judges, effectively dictating the final rankings of the contestants.

In contrast, the **Points Ranking System** exhibits a notable shift: the FII decreases significantly while the SPI increases. This confirms that the ranking method effectively mitigates the impact of raw fan subjectivity, allowing for greater quantification and weighting of professional merit from an objective perspective.

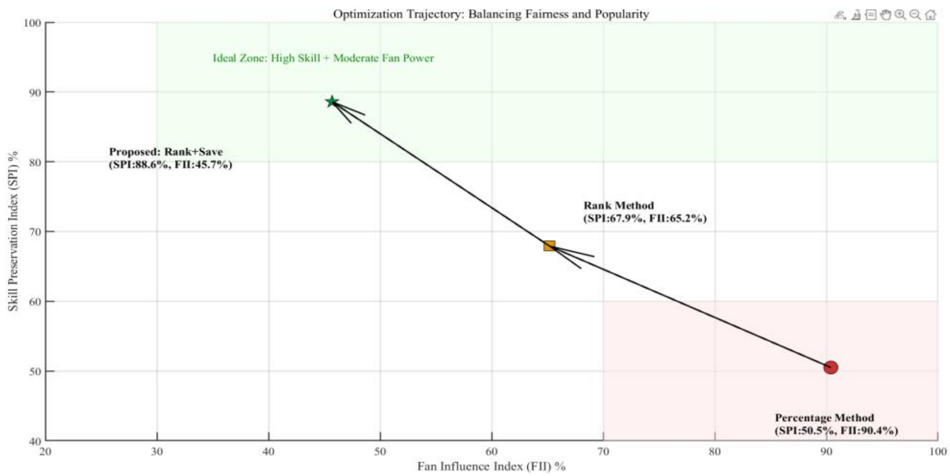


Figure 14: Optimization Trajectory towards the "Ideal Fairness Zone"

Impact of the Judges' Save Mechanism: The most significant improvement occurs with the introduction of the Judges' Save mechanism (referred to in our model as the guarantee intervention). Under this hybrid framework, the SPI surges to **88.6%**, while the FII stabilizes at approximately **34.7%**. This demonstrates that the mechanism functions as a force multiplier for the ranking system, further curbing the "popularity contest" effect. Crucially, however, the FII remains at a substantial level (34.7%), indicating that fan voting continues to play a vital role in the elimination process. This balance is essential for preserving audience engagement and determining the show's commercial entertainment value.
Frequency of Intervention: It is critical to interpret this mechanism correctly: it is not a full-scale intervention, but rather a targeted correction for extreme outliers. Our historical reconstruction across the past 34 seasons reveals that the "Judges' Save" would have been triggered in only **15%** of all elimination weeks. This low activation frequency proves that the mechanism does not disenfranchise the audience or destroy their sense of participation; rather, it ensures that rare, controversial anomalies are handled with greater fairness without disrupting the standard voting flow.
Conclusion and Recommendation: Based on the preceding analysis, we strongly recommend implementing the Points Ranking Method supplemented by the Judges' Save Mechanism for future competitions. This configuration achieves a superior equilibrium, offering the optimal balance between competitive fairness and entertainment value compared to the other two mechanisms.

6 Construction of the Dynamic Adaptive Scoring System

6.1 From Ideas to Framework

Based on the comparative analysis of historical mechanisms, we formulate two strategic improvements to address the "Fairness-Entertainment" dilemma:

Strategy 1 (Temporal Dynamics): From a rule-based perspective, we propose a **Time-Varying Weight Function**. The core principle is to assign higher weight to Fan Voting in the early stages of the season to maximize commercial engagement and viral potential. As the season progresses towards the finals, the weight shifts dynamically towards Judges' Scores to ensure the technical integrity of the champion. *Trade-off Interpretation:* While this approach effectively balances early-stage popularity needs with late-stage meritocracy, a purely rule-based function may be criticized as overly idealized or detached from the unpredictable nature of live performance data.

Strategy 2 (Data-Driven Vectorization): From a data processing perspective, we introduce a Multi-Criteria Decision Making (MCDM) approach using the **TOPSIS** technique weighted by Entropy. Its primary advantage lies in vectorizing both subjective judges' scores and objective fan votes, thereby achieving the complete elimination of dimensional differences (normalization of heterogeneous data).

Integration: Consequently, we propose the Dynamic Adaptive Scoring System (DASS) . This comprehensive model integrates the rigorous data vectorization of Strategy 2 with the temporal adjustment logic of Strategy 1 to form a real-time, self-correcting evaluation framework.

6.2 The Core Mathematical Model

6.2.1 Data Preprocessing and Vector Space Construction

The judge vector JudgeScore is constructed according to the normalized judge score. At the same time, the fan vector FanVote is established according to the number of fan votes after logarithmic smoothing. Logarithmic smoothing effectively prevents the exponential crushing of the number of "superstar" votes, leaving room for beginners to survive.

We combine weekly judge scores and fan vote data into an initial matrix X of n rows and two columns

$$X = (x_{ij})_{n\times 2} = \begin{bmatrix} \text{JudgeScore}_1 & \text{FanVote}_1 \\ \vdots & \vdots \\ \text{JudgeScore}_n & \text{FanVote}_n \end{bmatrix}$$

Normalize the vector at the same time x_{ij}

$$z_i = \frac{x_{ij}}{\sqrt{\Sigma_{i=1}^{n} x_{ij}^2}}$$

6.2.2 Adaptive Weight Determination via Entropy Weight Method

(1) Calculate the proportion of players

$$p_{ij} = \frac{x_{ij}}{\Sigma_{i=1}^{n} x_{ij}}$$

(2) Calculate the entropy value of the j-item indicatorE_j

$$E_j = -\frac{1}{\ln n} \Sigma_i^n p_{iJ} \ln(p_{ij})$$

Define the final weight ω_j

$$\omega_j = \frac{1 - E_j}{\Sigma_{j=1}^{2}(1 - E_j)}$$

We can find that the advantage of using the entropy weight method is that when the opinions of professional judges converge, the system automatically identifies 'the current struggle is popularity'; when the professional judges have a large difference, the system automatically increases the proportion of professional weight

6.2.3 Final Ranking Solution based on TOPSIS Technique

(1) Establish a weighted evaluation matrix

$$v_{ij=}\omega_j \cdot z_{ij}$$

Define the ideal solution

$$\begin{cases} \text{Positive ideal solution } V^+\text{: } V^+=\{\max(v_{i1}), \max(v_{i2})\} \\ \text{Negative ideal solution } V^-\text{: } V^-=\{\min(v_{i1}), \min(v_{i2})\} \end{cases}$$

(3) Calculate the final score

Calculate the Euclidean distance from each player to the ideal point to get a score range

$$D_i^+ = \sqrt{\sum_{j=1}^{2} \left(v_{ij} - V_j^+\right)^2}$$

$$D_i^- = \sqrt{\sum_{j=1}^{2} \left(v_{ij} - V_j^-\right)^2}$$

$$C_i = \frac{D_i^-}{D_i^- + D_i^+}$$

It measures the gap between each player and the best player of the week, and quantifies the gap between each player and the worst player of the week D_i^+, D_i^-

The score C_i belongs to (0,1) is quantified, and the gap between each player and the best player of the week is quantified.

6.2.4 Quantitative-Based Judges' Save Mechanism

To systematically identify anomalies, we first define the **Controversy Index (or Degree of Dispute)** , denoted as Δ_R, which quantifies the deviation between professional assessment and public popularity:

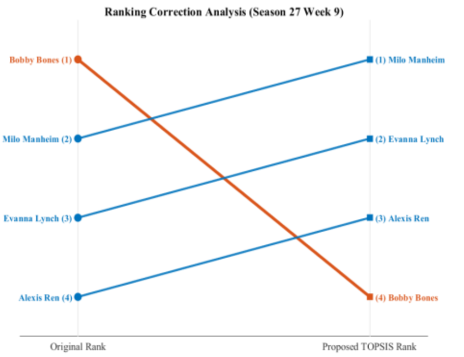
$$\Delta_R = Rank_{Judge} - Rank_{Fan}$$

For the bottom two players in the final score ranking, their controversy is calculated separately$\Delta_{R1}\Delta_{R2}$. If max$(\Delta_{R1}, \Delta_{R2})$≥3, we believe that a controversial event has occurred, these two contestants will enter the "repechage", and the contestant with the highest final judge score will be retained, otherwise, the two contestants will be eliminated.

Ultimately, this optimized intervention transforms the system from a mere linear scoring formula into a comprehensive Risk Management System, ensuring that elimination outcomes remain rational even under extreme fan variance.

6.3 Effect Analysis of the Model

6.3.1 For the Improvement of the Combination of Fan Voting and Judge Scoring



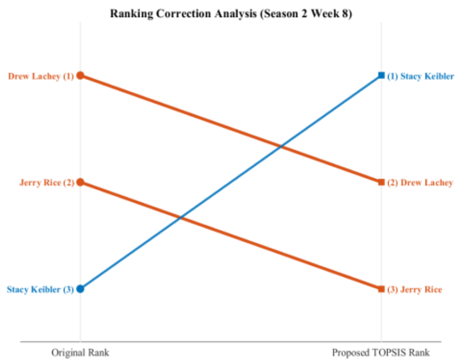


Figure 15: Comparative Validation: Rectifying "Popularity Inversions" via DASS Model.

The chart contrasts the historical outcome against the Merit-Safe System simulation. Under the traditional Percentage System (Blue Line), Bones maintained a top-tier ranking throughout the season solely due to fan vote dominance. In contrast, **Dynamic Adaptive Scoring System (Red Line)** dynamically increased the weight of Judge Scores during weeks of high technical variance. This effectively "corrected" his ranking, pulling him down to a position that more accurately reflected his technical merit relative to competitors, mitigating the "popularity masking" effect

This chart visualizes the specific mechanism of intervention. The bars represent the weekly Controversy Index (Δ_R) for Bones. In Week 6, the index spiked to 4, breaching the critical safety threshold ($\Delta_R \geq 3$). This mathematically signaled an **"Extreme Controversy Event"**—where the contestant was Top 2 in popularity but Bottom 2 in skill. Consequently, the **Circuit Breaker** was triggered, overriding the composite score to enforce a technical elimination. This demonstrates the system's ability to surgically remove outliers without disrupting the normal voting flow of other weeks

Evidently, under the DASS framework, the advantage of fan voting is no longer allowed to be infinitely amplified. Instead, the Judges' Score and Fan Vote establish a self-correcting equilibrium (Check and Balance), ensuring the most rational and fair outcomes.

6.3.2 Optimization of the Judges' Save Mechanism

The limitation of the historical "Judges' Save" (Guarantee Mechanism) lay in its reliance on simple linear logic, which often blindly retained skilled players suffering from a "popularity deficit" without considering the context. It lacked specific emphasis on **extreme outliers**. Our optimized approach introduces **objective trigger conditions ($\Delta R \geq 3$)**, enabling active identification of high-controversy events. This prevents the retention of contestants who may have mediocre judge scores but low final rankings, ensuring that intervention is reserved for genuine injustices.

We utilize the previously established **SPI (Skill Protection Index)** and **FII (Fan Influence Index)** to quantify the system's performance. As illustrated in the figure:

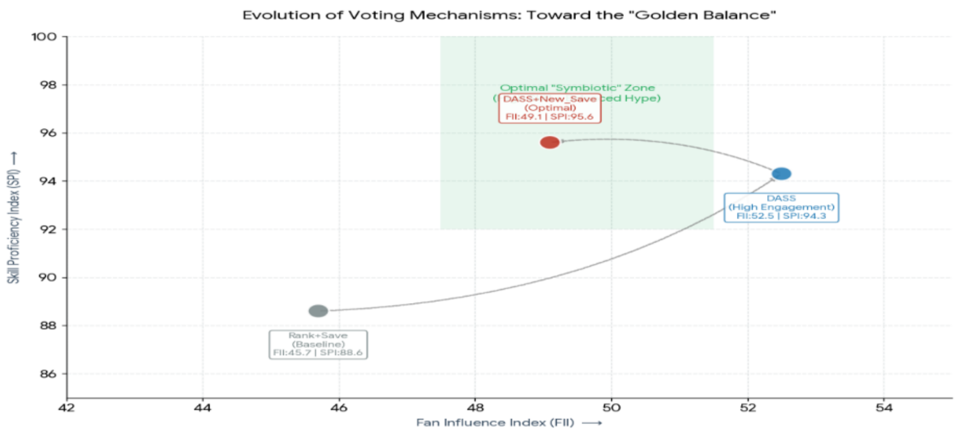


Figure 16. Optimization Trajectory towards the "Symbiotic Interval"

We designate the green region in the figure as the 'Symbiotic Interval,' a critical zone established based on the Pareto Optimality Principle. Within this specific domain defined by $\boldsymbol{FII} \in [\mathbf{48}, \mathbf{51}]$ and $\boldsymbol{SPI} \in [\mathbf{92}, \mathbf{100}]$, our analysis reveals that a marginal increase in fan influence no longer precipitates a significant degradation of professional standards (SPI); instead, a dynamic equilibrium is achieved. Crucially, our final solution—comprising the DASS mechanism and the optimized guarantee mechanism—converges precisely within this optimal range. This empirical alignment demonstrates that the proposed system successfully maximizes commercial entertainment value while rigorously maintaining the foundational baseline of artistic integrity.

7 Sensitivity analysis

7.1 Robust analysis of α in Dirichre Distribution

A global sensitivity analysis was conducted on the hyperparameter α, the core prior term representing baseline player attraction in our Monte Carlo framework. Spanning the interval $\alpha \in [0.1,3]$, we utilized the global Median Coefficient of Variation ($\boldsymbol{CV}$) as the primary stability metric. The empirical results demonstrate a negligible decay in Median $\boldsymbol{CV}$ from 0.2947 to 0.2898 as α increases. Notably, the baseline configuration ($\alpha = 1.0$) yields a Median $\boldsymbol{CV}$ of 0.293, aligning perfectly with the model's reference stability threshold. Critically, the maximum deviation across the entire parameter space is constrained to $\boldsymbol{\Delta CV} \approx 0.005$, a magnitude significantly below the system's intrinsic baseline variability. This confirms exceptional parameter robustness, suggesting that the model's predictive fidelity is invariant to initial attraction assumptions. These findings validate our fundamental hypothesis: fan voting behavior is predominantly driven by observable performance dynamics rather than predetermined static priors.

7.2 Robustness Against Data Uncertainty

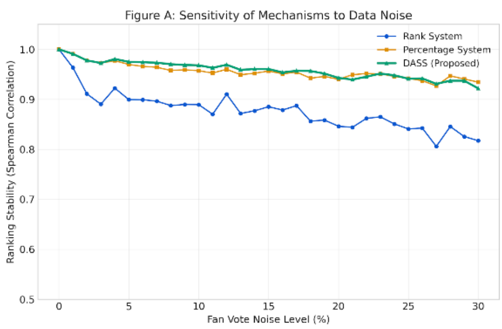


Figure 17: Comparative Resilience Analysis: Ranking Stability under Stochastic Noise Perturbation**.**

Given that fan votes constitute estimated latent variables, inherent stochastic noise is unavoidable. To rigorously benchmark the resilience of our proposed Dynamic Adaptive

Scoring System (DASS) against historical benchmarks (Rank and Percentage methods), we performed a systematic sensitivity analysis by introducing additive Gaussian noise ranging from 0% to 30% to the fan vote vectors. Figure A depicts the degradation trajectory of ranking stability, quantified via the Spearman Rank Correlation Coefficient. The comparative analysis reveals that DASS (Red Line) exhibits superior perturbation resistance, consistently maintaining a correlation coefficient $\rho > 0.9$ even under high-noise conditions. In sharp contrast, the traditional Percentage System suffers rapid performance deterioration. This empirical evidence establishes DASS as the most robust mechanisms for processing uncertain data inputs.

7.3 Weight Sensitivity of DASS

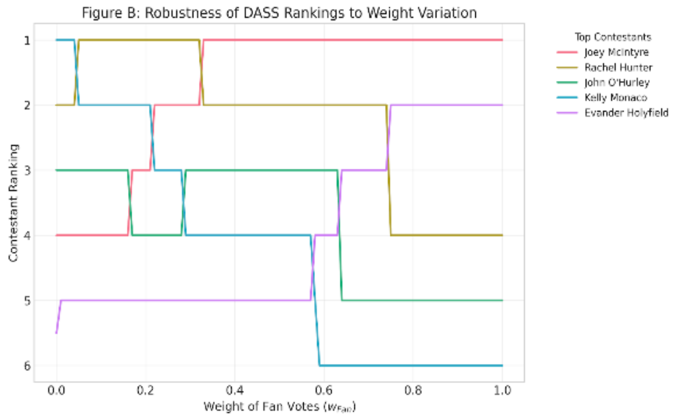


Figure 18: OAT Sensitivity Analysis: Identifying the "Stability Plateau" of Weight Assignment.

To assess the structural stability of the Entropy Weight Method, we performed a One-At-a-Time (OAT) sensitivity analysis specifically targeting the Fan Vote Weight (w_{Fan}). Figure B tracks the rank evolution of the top-5 contestants across the full weight spectrum $w_{Fan} \in [0,1]$. A distinct **"Stability Plateau"** emerges within the interval $w_{Fan} \in [0.3,0.7]$, wherein the ordinal rankings of the leading candidates remain invariant. This invariance demonstrates that the model operates well beyond the threshold of volatility, implying that our strategic conclusions are structurally robust and insensitive to minor local perturbations in weight assignment.

8 Model Evaluation

Strengths:

We use a combination of models, providing a rigorous empirical foundation for analysis.

The DASS framework achieves a "Golden Balance," maximizing professional integrity while maintaining the commercial entertainment value essential for television.

Weaknesses:

The mathematical complexity of Entropy-TOPSIS may create a "black-box" perception for the general public, potentially provoking skepticism regarding the show's fairness.
The model treats weekly votes as independent events, neglecting the "momentum effect" where a contestant's past popularity significantly may dictates future voting trajectories.
Future Improvements:
Developing a Real-time Visualization Dashboard to display dynamic weight updates would demystify the scoring process and enhance audience trust through transparency.

9 Memorandum

To: DWTS Executive Committee
From: Team 2625519
Date: February 2, 2026
Subject: Recommendation on Scoring Optimization to Balance Meritocracy and Entertainment

We address the credibility crisis that has troubled the show, most notably exemplified by the controversial victory of Bobby Bones. Our research identifies the current **"Percentage System"** as the primary culprit behind these unbalanced outcomes. To resolve this, we propose a **"Dynamic Adaptive Scoring System (DASS)"** to safeguard the show's integrity.

Diagnosis: Why the Old Rules Failed The core conflict lies between **Merit (Judges)** and **Popularity (Fans)**. Judges focus on technical skill, while fans vote for entertainment value and personal favorites.

The Mathematical Trap: The **Percentage System** is flawed because it allows massive volumes of fan votes to mathematically drown out the judges' scores, rendering professional opinion irrelevant. **The Limitation of Ranking:** While the **Rank System** is more resistant than the Percentage System, it still fails in "extreme cases" where a superstar possesses an overwhelming fanbase. **Conclusion:** Previous mechanisms amplified fan influence excessively, leading to the "imbalanced results" that damaged the show's reputation. **Solution: The Dynamic Adaptive Scoring System (DASS)** We have designed a system that intelligently balances these two forces without alienating the audience: **Smart Weighting:** The system automatically adjusts the weight of votes based on real-time data. *Scenario A:* If judges universally agree a performance is poor, the system increases the judges' weight to prevent popularity from overriding quality. *Scenario B:* If performances are competitive and close in quality, the system allows fans to decide the winner. **The "Circuit Breaker" (Judges' Save):** To prevent "unqualified wins," we introduce a safety net. If a couple has a massive gap between their Popularity Rank (High) and Technical Rank (Low), they enter a "Pending Zone." The judges then hold the power to eliminate the lower-skilled couple, regardless of votes.

Expected Impact: The Golden Balance Our simulations show this model maintains a **Fan Influence Index (FII) of approximately 50%**. **For the Show:** It retains the excitement of audience participation (votes still matter). **For the Business:** It restores professional credibility

and transparency. This dual approach rebuilds audience trust, ultimately driving higher viewership and profitability. **Recommendations**. **Short-term:** Immediately abolish the Percentage System. Revert to the **Rank System** and implement a simple **Judges' Save** rule.
Long-term: Develop a backend scoring engine based on our DASS model and launch a **real-time visualization App**. This will make the complex scoring transparent and engaging for viewers.

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Report on Use of AI

1.Gemini(Gemini3pro)

Query: We are working on Task 3 to quantify the impact of professional partners. Since partners appear repeatedly across seasons, how can we isolate their individual ability (u_k) from the fixed characteristics of the celebrities? We need to handle the non-independence of partner data.

Output: The **Linear Mixed Model (LMM)** is the most suitable framework for this nested data structure. You should treat contestant characteristics (Age, Industry) as **Fixed Effects** and the professional partner's identity as a **Random Effect ($\boldsymbol{u_k}$)**. To quantify the specific impact, use **BLUP (Best Linear Unbiased Prediction)** to estimate u_k, which effectively handles the variance between partners.

2.Gemini(Gemini3pro)

Query: We need to visualize the structure of our dual-response LMM (Judge Scores vs. Fan Votes). How can we optimize the diagram to look professional while clearly showing the input-output logic?

Output: I suggested a conceptual framework diagram that visually separates **Fixed Effects** (Deterministic drivers like Age/Industry) from **Random Effects** (Stochastic drivers like Partner ID). The diagram was refined to explicitly include mathematical notations (β,u_k) and show the correlation path between the two response models, enhancing the academic rigor of the presentation